

Institute for Energy Technology (IFE)
Instituttveien 18, 2007 Kjeller, Norway

August 5, 2026

Editorial Office
Computer Methods in Applied Mechanics and Engineering

Dear Editor,

We submit **“PDE-Agents: An LLM-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning”** for consideration as an original research article in CMAME.

The manuscript introduces a computational framework in which three specialist large language model agents autonomously set up, run, analyse, and iteratively improve finite element simulations from natural-language descriptions. A GraphRAG knowledge base (Neo4j, 768-dim vector embeddings) grounds agent reasoning in curated material properties, failure patterns, and prior run lineage. We propose a novel *KG Smart* integration strategy and show that *integration pattern*—not knowledge content—determines whether retrieval augmentation helps or hinders an LLM agent.

The contribution is methodological: we present and rigorously evaluate a new computational framework, not merely apply an existing solver. The evaluation includes V&V confirming $\mathcal{O}(h^2)$ convergence, a three-way ablation across 50 tasks, a novel-material experiment (MPF = 1.00 vs. 0.34 baseline), and production metrics from 1,369 runs (97.8% success). All code and data are openly available at <https://github.com/MatPro-IFE/pde-agents>.

This work falls within CMAME’s scope of physically based machine-learning methods and computational automation. No prior work combines multi-agent LLM orchestration with production FEM solvers and graph-based retrieval for computational mechanics. The manuscript is original, not under consideration elsewhere, and all authors have approved its submission.

Sincerely,

Sayan Adhikari
Institute for Energy Technology
(IFE), Kjeller, Norway
sayan.adhikari@ife.no